* LEDs →

Intersection 1

Pin	Leg		
R → 22	→ LED +	−	LED → GND
Y → 23	→ LED +	−	LED → GND
G → 24	→ LED +	−	LED → GND

Intersection 2

R → 25	→ LED +	−	LED → GND
Y → 26	→ LED +	−	LED → GND
G → 27	→ LED +	−	LED → GND

Intersection 3

R → 28	→ LED +	−	LED → GND
Y → 29	→ LED +	−	LED → GND
G → 30	→ LED +	−	LED → GND

Intersection 4

R → 31	→ LED +	−	LED → GND
Y → 32	→ LED +	−	LED → GND
G → 33	→ LED +	−	LED → GND

* Ultrasonic sensors

Intersect: 1 :

Sensor	→	Mega Pin
Vcc	→	5V
GND	→	GND
TRIG	→	2
ECHO	→	3

Intersect: 2 :

Sensor	→	Mega Pin
Vcc	→	5V
GND	→	GND
TRIG	→	4
ECHO	→	5

Intersect: 3 :

Sensor	→	Mega Pin
Vcc	→	5V
GND	→	GND
TRIG	→	6
ECHO	→	7

Intersect: 4 :

Sensor	→	Mega Pin
Vcc	→	5V
GND	→	GND
TRIG	→	8
ECHO	→	9

* Switches

* All_red :

Switch	→	Mega
Leg	→	10
Leg	→	GND

* Intersection 1

Switch	→	Mega
Leg	→	11
Leg	→	GND

* Intersection 2

Switch	→	Mega
Leg	→	12
Leg	→	GND

* Intersection 3

Switch	→	Mega
Leg	→	13
Leg	→	GND

* Intersection 4

Switch	→	Mega
Leg	→	14
Leg	→	GND

⚠ all switches are active low (Input_Pullup)

⚠ * the lights should be connected to the same GND

* Switches & Sensors should also be connected to the same sensor

* Might need low value Resistors (10Ω – 300Ω) for each LEDs.